A3-PS003-TC520 · Chauraha Sync

Federated Traffic-Flow Prediction
with Communication-Cost Evaluation

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AI-assisted cross-checks · October 19, 2026

Problem Description

Can local traffic forecasting retain accuracy while reducing accounted data exchange?

Measure centralised, local-only and FedAvg training under random IID-style and quiet-to-busy non-IID sensor groups.

5 clients · 34 sensors/client · 3 seeds
15 primary learned runs; 6 extended-round diagnostic runs.

Public PEMS08 data; next five-minute forecast. No live cameras or city deployment.

AI-assisted checks — primary results

Centralised MAE: **13.862 ± 0.013**
IID FedAvg: **13.964 ± 0.014** (+0.73%)
Non-IID FedAvg: **14.209 ± 0.066** (+2.50%)

Common-group relative-error gap: **2.404 → 2.919 percentage points** for IID versus non-IID FedAvg.

Quietest group: non-IID FedAvg MAE **10.42% worse** than matched local-only. H4 is refuted.

Mean ± sample SD over three seeds. Primary: 30 central/local epochs; 30 FedAvg rounds × 1 local epoch.
Counts and figures derive from frozen JSONs, not the superseded Level 1 results.

Background

Public benchmark, not Bengaluru deployment

PEMS08 tensor: **17,856 × 170 × 3**.
Five-minute steps; channel 0 is flow.
California freeway sensors, grouped into five simulated clients. [2, 3]

Quality and modelling assumption

NaN = **0**; Inf = **0**.
Zero flow: **10,560 entries · 0.348%**.
Treat exact zero as missing; drop a window if any of its 13 flow values is zero.

Leakage-safe chronological protocol

Train/validation/test: **70/15/15%**.
Eligible windows: **2,098,136 / 448,263 / 448,533**.
Train-only scaling; no cross-boundary windows.
Validation selects checkpoints.

Measured heterogeneity

IID-style sensor-mean W1, seeds 0/1/2:
28.407 / 30.085 / 30.793.
Quiet-to-busy non-IID: **127.867**.
Primary statistic: pairwise W1 on sensor means.

Zero handling is an explicit assumption, not proof about every zero. W1 uses valid training values only.

Literature & RFC Review

[1] McMahan et al., 2017 — FedAvg
Iterative averaging of locally trained model parameters. We implement sample-weighted aggregation with all five clients; the local optimiser is Adam.

[2, 3] Caltrans PeMS and the ASTGNN benchmark package
Public California traffic measurements and the exact PEMS08 tensor. We use the dataset, not ASTGNN's graph model; our forecaster has only 513 parameters.

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[4] AIORI PS-003 — measurable comparison, not a new algorithm
Contribution: matched local-only references, identical evaluation groups, validation-selected checkpoints, three-seed reporting and explicit 30-versus-100-round payload trade-offs.

No governing FedAvg RFC is claimed. No differential privacy, secure aggregation or transport-layer bandwidth measurement is implemented.

Proposed Solution

Frozen NumPy scientific core

12 flow lags + sin/cos of global index mod 288.
14 → 32 ReLU → 1; 513 parameters.
MSE; Adam, LR 0.001; batch 1,024.

FedAvg + matched comparisons

5 clients; sample-weighted parameter average.
Local Adam state resets each round.
Central/local: 30 epochs; FedAvg: 30 rounds × 1 local epoch. Seeds 0, 1, 2.
Minimum validation MAE selects checkpoints.

Working demonstration prototype

Local PEMS08 file → eligible held-out sensor window → real saved-weight forecast.
Recorded truth + persistence comparison.
Browser SHA checks; NumPy parity tolerance 0.001 flow units. No retraining in this view.

Inspect the evidence and cost

Three-seed metrics; common-group errors; verified validation-history replay.
2,052 B/model; **20,520 B/round**.
Training-flow reference: **8,499,320 B**.

Browser replay is not training; fresh NumPy smoke commands write separate outputs. No city deployment or privacy guarantee.
Shared train-derived scaling is a simulation assumption. Equal sample passes are not equal optimiser trajectories.
Payload excludes wire/serialization overhead; zero model exchange does not mean zero total system cost.

Optimization Proposed — Accuracy & Client Cost

Primary: 30 central/local epochs; 30 FedAvg rounds. Test means ± sample SD, seeds 0/1/2.

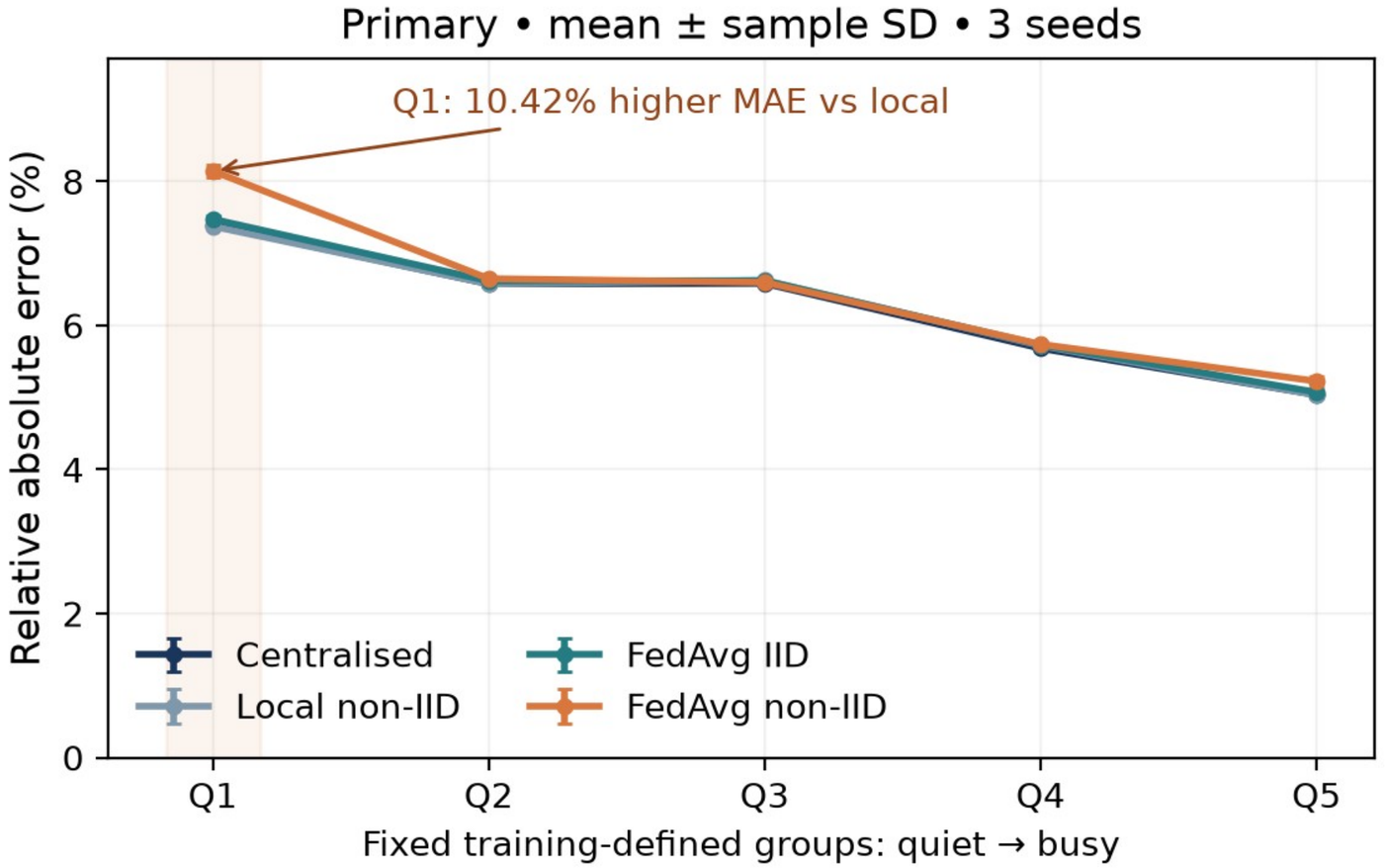
Method	Test MAE	Test RMSE
Persistence	16.078	24.451
Daily / sensor	34.776	51.078
Centralised	13.862 ± 0.013	21.074 ± 0.008
Local IID	13.992 ± 0.031	21.221 ± 0.048
Local non-IID	13.898 ± 0.005	21.096 ± 0.012
FedAvg IID	13.964 ± 0.014	21.211 ± 0.031
FedAvg non-IID	14.209 ± 0.066	21.373 ± 0.079

FedAvg: 615,600 model-payload B/run.
Local-only: 0 model-exchange B, not total cost.

Separate diagnostic: 30 → 100 rounds

IID MAE	13.964 → 13.873 ± 0.010
Non-IID MAE	14.209 → 14.064 ± 0.043

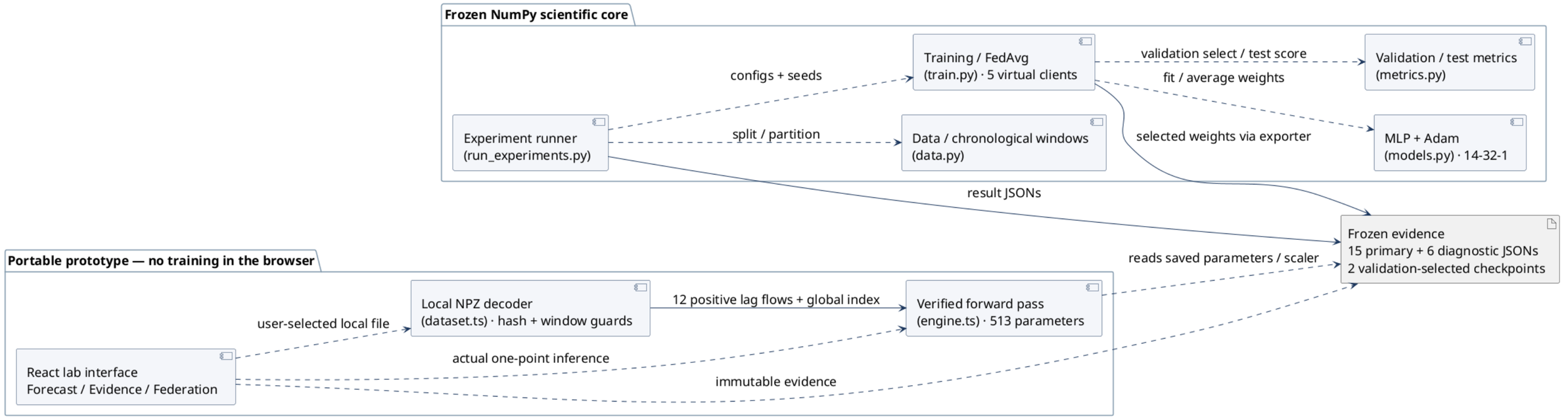
2,052,000 B/run · 3.33× payload; no plateau claim.



Q1: 7.983 vs 7.230 MAE; H4 refuted. Scarcity benefit remains untested.
Relative error = group MAE / group mean flow, not MAPE.

Solution Architecture & Design

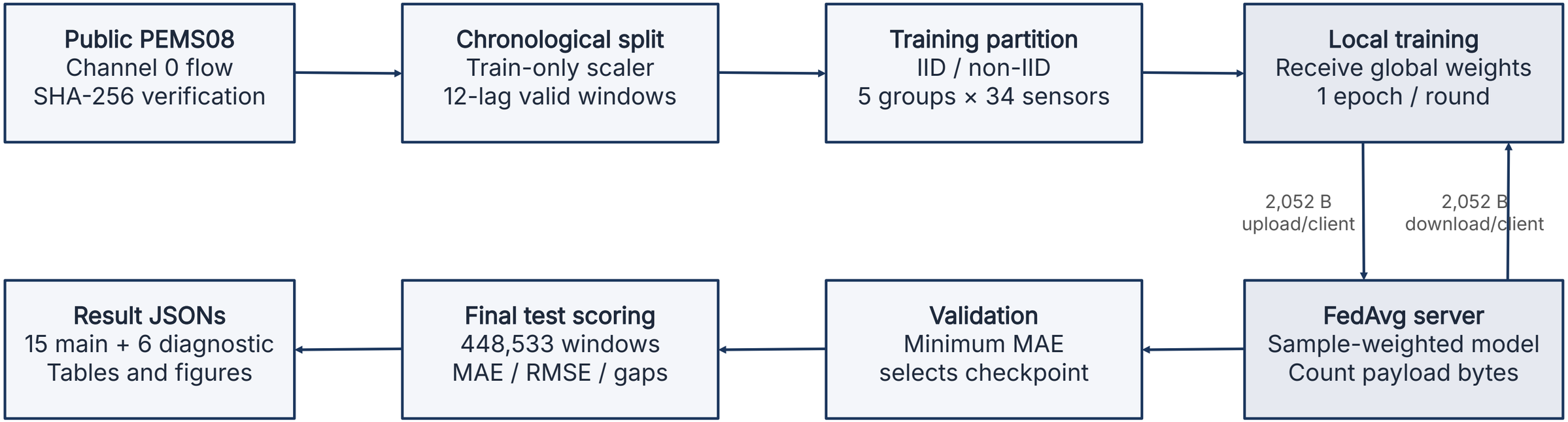
Actual module dependencies: frozen NumPy simulator + a portable, checksum-verified browser inference view.



Dashed arrows: module dependencies. Solid arrows: result/model artifacts and validated input flow.
Browser playback is not training. Local file input is not an app upload; no deployed city or formal privacy claimed.

Data Flow Diagram

No test-driven selection. Local training uses training windows; validation is evaluated after aggregation.



Rounds to validation target, seeds 0 / 1 / 2: IID 3 / 2 / 3; non-IID 9 / 6 / 4.
Target = 1.05 × each seed's best centralised validation MAE; descriptive, not a stopping rule.
Five clients × two directions × 2,052 B = **20,520 B/round**.
Figure: simulation data flow; no production network or formal privacy claim.

Deployment Timeline — Technical Release & Review

Technical artifacts prepared October 19. Deployment here means the local simulator release, not a city installation.

Date / status	Milestone	Evidence / exit check
Oct 19 · complete	Dataset, chronological windows and baselines	Hashed data; training-only statistics
Oct 19 · complete	Primary matrix and separate extended diagnostic	15 + 6 JSONs; three seeds
Oct 19 · complete	Actual inference, reports and evidence figures	2 verified saved checkpoints; runnable predictor
Oct 19 · complete	Release verification and AIORI-format presentation	55 tests; 41 preserved entries; PDF/PPTX
Oct 19–30 · review	Team reproduction, mentor walkthrough, publication	Record actual checks and feedback; team uploads
Oct 31 · window end	Official Level 2 submission window closes	Confirm final repository access and artifacts

Mentoring dates will follow organiser confirmation; session dates and feedback will be logged when held.
October 30 is our buffer target. Later review/publication dates are not claimed as new experiment dates.

Bibliography

[1] McMahan, B., Moore, E., Ramage, D., Hampson, S. and Agüera y Arcas, B. (2017). Communication-Efficient Learning of Deep Networks from Decentralized Data. AISTATS, PMLR 54:1273–1282. proceedings.mlr.press/v54/mcmahan17a.html

[2] California Department of Transportation. Performance Measurement System (PeMS) Data Source. dot.ca.gov/programs/traffic-operations/mpr/pems-source

[3] Guo et al. ASTGNN code and benchmark data repository. Exact PEMS08 tensor independently hashed for this study. github.com/guoshnBJTU/ASTGNN

[4] AIORI-3. Official PS-003 brief, presentation template and hackathon guideline. portal.aiori.in/aiori-3/hackathon-guideline/

[4] Supplied PDFs are preserved in docs/PS-003_official.pdf and docs/Proposed-structure-hackathon.pdf.
The URL links the public guideline, not those PDFs. URLs checked October 19; metrics come from preserved JSONs.
Public availability is not treated as a dataset redistribution licence; obtain the benchmark from its source.

Thank You for Your Attention!

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Repository
github.com/jhaaryan7580-dev/A3-PS003-TC520

Introduction video
youtube.com/watch?v=jJh9rvyaiUA

LinkedIn: linkedin.com/in/aryan-jha-785a29343

Repository QR
Private access may require login.
Mentor/evaluator access must be checked.

Reproducible evidence, honest limitations and mentor feedback guide the final submission.